

100 Days of Azure DevOps

Day 88 Handout — The cluster stops being a petting zoo for kubectl

9 - Advanced & Enterprise · Personal lab only · Educational content · Not a sales pitch

Architecture A — push CD vs pull GitOps

Dimension	Azure Pipelines push (Day 57)	Flux / Argo pull
Source of truth	Pipeline run + what it last applied	Git commit the controller wants
Who talks to the API	Pipeline job (service connection / kubeconfig)	In-cluster controller reconciles
Change window	YAML stage + env approval	PR review, then merge; cluster follows
Drift	Unless you re-run, drift can hide	Reconcile loop notices and heals or fights you
Hotfix temptation	az / kubectl in a job or from a laptop	kubectl apply is debt; PR is the path
Lab default	Push is enough for App Service / Container Apps	GitOps when a cluster is the product

Architecture B — when to pull, when to keep pushing

Workload	Default	Move to GitOps when
App Service / Functions zip	Push from Azure Pipelines	Never required
Container Apps revision	Push image tag/digest from CI	You want the revision YAML in Git as truth
AKS / any cluster	Optional local Flux quickstart	Multiple people kubectl and the zoo starts
Secret objects	Sealed-secrets / External Secrets / KV CSI	Plain Secret YAML in a public repo
No cluster today	Written comparison only	Do not buy AKS just to install Flux

Architecture C — petting-zoo failure modes

Pitfall	What you see	Fix
Cowboy kubectl	Three people apply; Git is a rumor	If it is not in the repo, it will drift back
Pipeline push + silent kubectl	CI green, cluster different	Pick a source of truth; disable the habit
Latest tag in GitOps	Controller 'reconciles' a moving target	Immutable digest / Build.BuildId
Secrets in the GitOps repo	Public fork has the connection string	External Secrets / KV; never raw Secret YAML
Flux on a forgotten AKS	Controller + cluster bill after the lesson	Local kind/minikube or paper only; destroy AKS

One-liner to remember

GitOps means the cluster stops being a petting zoo for kubectl.

Step-by-step lab (personal Azure / Azure DevOps only)

Read first. Optional local Flux. Personal Azure only — destroy any AKS you start. No employer cluster kubeconfig.

- 1 Read the GitOps principles (desired state in Git, declarative, pulled, continuously reconciled). Write them in docs/gitops-day88.md.
- 2 Write a push-vs-pull table comparing Azure Pipelines apply (Day 57) to Flux/Argo reconcile.
- 3 Optional: Flux quickstart on local kind/minikube — not on a leftover paid AKS unless you will destroy it tonight.
- 4 Add a sample manifests/gitops/kustomization.yaml (or Flux GitRepository YAML) that points at this personal repo path. Do not apply to a work cluster.

- 5 Write the cowboy rule: no kubectl hotfix habit even in a lab. A PR changes desired state.
- 6 If you skip the cluster, the written comparison still finishes the lab. Say so in the markdown.

Desired state fragment — a controller would reconcile this, not your laptop

```
# manifests/gitops/kustomization.yaml (sample; apply only to a personal local cluster)
apiVersion: kustomize.config.k8s.io/v1beta1
kind: Kustomization
resources:
  - deployment.yaml
images:
  - name: demo-shipboard
    newName: <youracr>.azurecr.io/demo-shipboard
    newTag: "$(Build.BuildId)" # pin a digest in real use; never latest-as-truth
# PR to change prod. No kubectl cowboy moves.
```

Done checklist

- Push-vs-pull written in docs/gitops-day88.md
- Sample manifest or Flux YAML in the personal repo
- No work-cluster kubeconfig
- Any AKS started for this lab is destroyed, or you used local-only
- Posted the LinkedIn document

Tomorrow — Day 89

Scaling DevOps for teams — five paved-road capabilities, not a queue that only says no